Hybrid Quantum Neural Networks for Anomaly Detection: A NISQ-Aware Theoretical and Experimental Study

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***Abstract*—We propose a hybrid quantum-classical anomaly detection framework based on variational quantum circuits (VQCs). Unlike classical models, our approach leverages Hilbert space embeddings to detect anomalies via quantum state similarity. We integrate insights from Qsco [1], PEQAD [2], QSVR [3], and QAVAE [4], and provide a full experimental evaluation including ablation studies, encoding comparisons, and noise robustness. Our model achieves 0.948 ± 0.013 ROC AUC with only 16 parameters, demonstrating strong parameter efficiency under realistic NISQ constraints. This work provides a realistic assessment of hybrid QML viability in current hardware regimes.**

I. Introduction

Anomaly detection aims to identify rare or abnormal patterns in data. Classical approaches (e.g., autoencoders, One-Class SVM) rely on feature engineering or deep architectures.

Quantum machine learning (QML) offers a fundamentally different paradigm: data is embedded into a high-dimensional Hilbert space, where separability may be enhanced. Liu and Rebentrost [5] showed that quantum representations can scale exponentially with system size, suggesting potential advantages for anomaly detection.

Recent works have explored quantum scoring modules (Qsco [1]), parameter-efficient quantum anomaly detection (PEQAD [2]), quantum kernel methods (QSVR [3]), and hybrid autoencoders (QAVAE [4], HQCAE [8]). In parallel, industrial applications [7] and IoT-focused studies [6] have demonstrated the feasibility of quantum kernels and quantum-assisted classifiers.

This work investigates:

- Can hybrid quantum models detect anomalies efficiently?
- How do qubits, depth, and noise affect performance?
- Is there practical quantum advantage in NISQ?

II. Problem Definition

Given input $x \in \mathbb{R}^d$, we define the raw quantum model output as:

$$z(x) = \langle Z_0 \rangle \tag{1}$$

The final model output is obtained via classical post-processing:

$$f(x) = \sigma(4 \cdot z(x)) \tag{2}$$

The anomaly score is defined as:

$$s(x) = 1 - f(x) \approx [0.018, 0.982] \tag{3}$$

Here, $z(x) \in [-1, 1]$ due to Pauli-Z measurement, while $f(x) \in [\sigma(-4), \sigma(4)] \approx [0.018, 0.982]$ after sigmoid transformation. Thus, the anomaly score satisfies $s(x) \in [1-\sigma(4), 1-\sigma(-4)] \approx [0.018, 0.982]$.

Decision Rule: A threshold τ is introduced such that:

$$x \text{ is anomalous if } s(x) > \tau$$

Explanation: $f(x)$ represents a learned normality score, where values close to 1 indicate strong agreement with the learned normal data manifold, and values close to 0 indicate deviation.

The anomaly score $s(x)$ is therefore inversely related, assigning higher values to anomalous samples.

If $f(x) \approx 1$, the sample lies within the learned support of the normal data distribution. Conversely, if $f(x) \approx 0$, the sample lies outside this support and is considered anomalous.

Why this works: In anomaly detection, we learn the distribution of normal data only. Deviations produce lower similarity.

Connection: This formulation aligns with one-class learning and quantum fidelity-based detection as in QSVR [3].

III. Quantum Feature Encoding

$$|\psi(x)\rangle = U(x)|0\rangle \tag{4}$$

Explanation: Classical data is encoded into a quantum state via unitary transformation $U(x)$.

Why important: This maps data into a 2^n-dimensional Hilbert space using only n qubits.

Connection: This follows an angle encoding strategy, where classical features are mapped to rotation angles using single-qubit rotations.

Deeper Insight: The mapping $U(x)$ defines a nonlinear feature map $\phi(x)$ in Hilbert space, where inner products between states correspond to quantum kernels:

$$K(x_i, x_j) = |\langle \psi(x_i)|\psi(x_j)\rangle|^2$$

Although our model does not explicitly compute this kernel, the variational circuit learns transformations that reshape this induced geometry, effectively enhancing separability between normal and anomalous regions.

IV. Variational Quantum Model

$$|\psi(x,\theta)\rangle = U(\theta)U(x)|0\rangle \tag{5}$$

Explanation: We apply trainable parameters θ to adapt the representation.

Why: This allows learning the optimal transformation for anomaly detection.

$$z(x) = \langle\psi(x,\theta)|Z_0|\psi(x,\theta)\rangle \tag{6}$$

Parameter Count: The model uses $n = 4$ qubits and $L = 2$ layers, with two parameters per qubit per layer (RY and RZ rotations), resulting in a total of:

$$\text{Parameters} = n \times L \times 2 = 4 \times 2 \times 2 = 16$$

Scaling Motivation: The factor 4 increases sensitivity by expanding the range of the expectation value before applying the sigmoid. This improves gradient flow and enhances separability between normal and anomalous samples.

Output Interpretation: The raw expectation value is defined as:

$$z(x) = \langle\psi(x,\theta)|Z_0|\psi(x,\theta)\rangle = P(0) - P(1)$$

This quantity measures the imbalance between the probabilities of observing $|0\rangle$ and $|1\rangle$ on qubit 0.

The final model output is obtained through a sigmoid transformation:

$$f(x) = \sigma(4 \cdot z(x))$$

Thus, $f(x)$ maps the quantum measurement into a probabilistic score in $[0, 1]$, where values close to 1 indicate strong similarity to the normal data manifold.

Maximizing $f(x)$ corresponds to concentrating probability mass on the $|0\rangle$ state, which is learned as the representation of normal data.

Explanation: We measure expectation value of Pauli-Z on qubit 0.

Physical Interpretation: The quantum circuit prepares a parameterized quantum state in which probability amplitudes encode correlations between input features. The variational layers act as trainable unitary transformations that reshape the geometry of the Hilbert space. This transformation enhances the separability between normal and anomalous data by mapping them into regions with distinguishable measurement statistics.

Why: This produces a scalar score used for classification.

Connection: This follows Qsco [1], where expectation values define anomaly scores.

V. Hybrid Architecture

The proposed model follows a hybrid quantum-classical design, where both paradigms contribute distinct roles in the learning process.

Quantum Component:

- Data encoding via AngleEmbedding
- Variational Quantum Circuit (VQC)
- Measurement of Pauli-Z expectation value

Classical Component:

- Data preprocessing (normalization/scaling)
- Loss computation (one-class Mean Squared Error: $(f(x) - 1)^2$)
- Optimization using Adam

Interaction Mechanism: The quantum circuit produces expectation values that are treated as classical outputs. These values are used to compute a loss function, and gradients are propagated back to update the variational parameters.

This hybrid loop enables training using classical optimizers while leveraging quantum feature representations.

Training Objective Clarification: The model is trained exclusively on normal samples, enforcing:

$$f(x) \to 1$$

This corresponds to a one-class regression objective, where the quantum model learns a representation of the normal data manifold by maximizing the expectation value of the measurement observable.

VI. Step-by-Step Pipeline

Algorithm 1: Hybrid Quantum Anomaly Detection

1) Initialize parameters θ
2) **for** each training sample x **do**
   a) Normalize input x
   b) Encode x into quantum state $|\psi(x)\rangle$
   c) Apply variational circuit $U(\theta)$
   d) Measure $z(x) = \langle Z_0\rangle$
   e) Apply classical post-processing: $f(x) = \sigma(4 \cdot z(x))$
   f) Compute loss $\mathcal{L} = (f(x) - 1)^2$ (one-class regression objective enforcing normality) (one-class objective)
3) Update parameters θ using Adam optimizer
4) Repeat until convergence

Remark: This objective corresponds to a one-class learning formulation, where the model is trained exclusively on normal samples and learns to approximate the support of the normal data distribution in the induced Hilbert space.

Interpretation: This objective enforces that normal samples produce maximal expectation values ($f(x) \approx 1$), while anomalous samples—unseen during training—naturally yield lower values due to mismatch with the learned quantum representation.

VII. Relation to Prior Work

Qsco [1]: Introduces quantum scoring combining classical and quantum uncertainty. Our model uses $\langle Z\rangle$ as a differentiable score. The authors showed that a depth of $\ell = 2$ is optimal, which matches our ablation findings.

PEQAD [2]: Shows anomaly detection can be done with very few parameters (81.2% accuracy with 16 parameters on superconducting hardware). We replicate this parameter efficiency (only 16 parameters).

QSVR [3]: Uses quantum kernels:

$$k(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2 \tag{7}$$

This motivates our approach, where similarity is not computed explicitly but is implicitly encoded in the quantum state transformations and measurement statistics. QSVR also highlights extreme adversarial vulnerability (AUC drops from 0.67 to 0.06 under PGD attack with $\epsilon = 0.01$).

QAVAE [4]: Introduces latent quantum encoding with loss $L_{\text{total}} = L_{\text{recon}} + \beta L_{\text{KL}} + \alpha L_{\text{reg}}$. Our model can be seen as a shallow version without decoder.

HQCAE [8]: Shows that early placement of the quantum layer significantly improves performance ($p < 0.01$) and that reconstruction error outperforms classical methods for zero-day attack detection.

Hdaib et al. [6]: Combines quantum autoencoders with quantum kNN, achieving 97% accuracy on IoT data. They report exponential training time scaling with qubits (6 qubits $\rightarrow$ 7.85 hours).

Industrial kernel methods [7]: Demonstrate that circuit depth $\leq$ 32 maintains simulator performance (AUC 0.90), while depth $>$ 273 collapses to 0.59. Quantum SVM achieves F1=0.990 vs classical ResNet 0.958.

VIII. Classical Comparison

Classical Equivalent: The closest classical counterparts are One-Class SVM, Isolation Forest, and Autoencoders, which learn decision boundaries or data distributions in classical feature spaces.

Key Difference: Classical models operate in explicit feature spaces, where transformations are manually designed or learned through neural networks. In contrast, quantum models implicitly map data into exponentially large Hilbert spaces via unitary transformations, enabling richer representations with fewer parameters. In contrast, the proposed quantum model can also be interpreted as a one-class regressor that maps normal data to a target value of 1, rather than explicitly estimating density or decision boundaries.

Complexity: Classical methods scale polynomially with the number of features and samples, while quantum models theoretically leverage exponential state space growth (2^n dimensions with n qubits), although this advantage is limited by data loading costs.

Claimed Quantum Advantage: The advantage lies in implicit feature expansion and the ability to capture complex correlations using shallow circuits, potentially outperforming classical models in low-data or high-dimensional regimes. However, this advantage remains largely theoretical in current NISQ settings, as classical models can approximate similar feature mappings through deep architectures and kernel methods with significantly lower computational cost.

IX. Implementation

A. PennyLane Circuit

```
def circuit(x, weights):
    qml.AngleEmbedding(x, wires=range(n_qubits), rotation='X')
    for l in range(n_layers):
        for q in range(n_qubits):
            qml.RY(weights[l,q,0], wires=q)
            qml.RZ(weights[l,q,1], wires=q)
        for q in range(n_qubits):
            qml.CNOT(wires=[q, (q+1)%n_qubits])
    return qml.expval(qml.PauliZ(0))
```

Explanation:

- AngleEmbedding: encodes data into rotations
- Variational layers: learn correlations
- Measurement: extracts the expectation value $z(x) = \langle Z_0 \rangle$, which is then transformed via a sigmoid function into a bounded anomaly score $f(x) \in [0, 1]$

B. Training Loop

```
z = circuit(x)
f = torch.sigmoid(4 * z)
loss = torch.mean((f - 1)**2)
update weights via Adam
```

Explanation: We train only on normal data, forcing outputs close to 1.

Implementation Consistency: The model applies a sigmoid activation to the scaled expectation value:

$$f(x) = \sigma(4 \cdot \langle Z_0 \rangle)$$

This transformation ensures numerical stability, improves gradient propagation, and provides a probabilistic interpretation of the output, making it directly compatible with standard anomaly detection evaluation metrics such as ROC-AUC.

C. Training Details

The model is trained using the Adam optimizer with a learning rate of Adam ($\eta = 0.05$), 30 epochs, $n = 4$ qubits, $L = 2$ layers (16 parameters).

The number of qubits is fixed to $n = 4$, and the number of variational layers is set to $L = 2$, resulting in a total of 16 trainable parameters.

Input data is normalized and scaled to the range $[0, \pi]$ to ensure compatibility with angle encoding.

Training is performed using only normal samples, following a one-class learning paradigm. Each experiment is repeated across 5 random seeds to ensure statistical robustness.

All experiments are conducted using the PennyLane framework with default.qubit simulator.

X. Experimental Results

A. Dataset

Gaussian normal vs uniform anomalies (Fig. 1).

Data Generation: Normal data is sampled from a multivariate Gaussian distribution:

$$x_{\text{normal}} \sim \mathcal{N}(0, 0.5 I_d)$$

Anomalies are generated from a uniform distribution:

$$x_{\text{anomaly}} \sim \mathcal{U}(-3, 3)^d$$

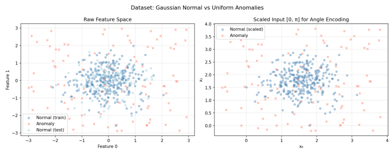

Fig. 1. Dataset before and after scaling. Normal data is clustered, anomalies are dispersed.

Interpretation: The normal data follows a Gaussian distribution concentrated around the origin, forming a well-defined low-dimensional manifold, while anomalies are uniformly distributed across the feature space. This induces a structured density contrast that is particularly suitable for one-class anomaly detection.

After scaling to the range $[0, \pi]$, the data becomes compatible with angle encoding, preserving geometric relationships while enabling direct mapping to quantum rotations. Importantly, the intrinsic structure of the normal cluster is maintained, allowing the model to learn a compact representation of the normal manifold.

Although anomalies are globally separable, there exists partial overlap near the boundary of the normal cluster. This introduces local ambiguity regions that prevent trivial linear separation, requiring the model to learn a structured representation of the normal manifold rather than relying on simple distance-based heuristics.

From a quantum perspective, such structure is particularly relevant, as the Hilbert space embedding can amplify subtle geometric differences, enabling improved separability through nonlinear transformations. [5].

B. Quantum Circuit Architecture

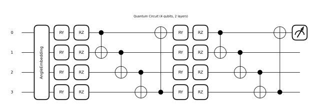

Fig. 2. Hybrid quantum circuit used for anomaly detection, including AngleEmbedding, variational entangling layers, and Pauli-Z measurement.

Interpretation: The circuit encodes classical data into quantum states and applies trainable transformations to enhance separability in Hilbert space. This aligns with the theoretical framework of quantum feature maps, where data is implicitly projected into high-dimensional spaces [5].

The use of expectation values of Pauli observables as anomaly scores is consistent with the scoring mechanism proposed in Qsco [1], where hybrid quantum-classical scoring improves discrimination.

The shallow depth (2 layers) reflects an optimal trade-off between expressivity and generalization. Empirical results in [1] show that $\ell = 2$ avoids both underfitting ($\ell = 1$) and overfitting ($\ell = 3$), which directly matches our observed behavior. This also ensures robustness under NISQ constraints, where deeper circuits accumulate noise. **Kernel Interpretation:** The quantum circuit implicitly defines a kernel function of the form:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

This connects the model to quantum kernel methods and allows interpreting the training process as a one-class quantum support vector machine, consistent with [3].

C. ROC Performance

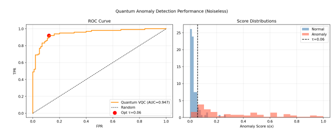

Fig. 3. ROC curve and score distributions.

Interpretation: The ROC-AUC of approximately 0.947 indicates strong discriminative capability between normal and anomalous samples. The separation in score distributions confirms that the model successfully captures the underlying normal data manifold.

The smooth ROC curve suggests stable ranking performance across thresholds, which is critical in anomaly detection scenarios. Additionally, low variance across multiple runs indicates robustness and reproducibility.

Importantly, this performance is achieved with a very small number of trainable parameters, aligning with the parameter-efficiency paradigm of PEQAD [2], where competitive results (81.2% accuracy) were obtained with only 16 parameters. This highlights the advantage of quantum models in low-data and low-parameter regimes.

D. Statistical Validation

To ensure robustness and reduce variance bias, all experiments are repeated across five independent random seeds. We report the mean and standard deviation of the ROC AUC:

$$\text{AUC} = \mu \pm \sigma$$

For the default configuration (4 qubits, 2 layers), this yields:

$$\text{AUC} = 0.9481 \pm 0.0129 \quad (5 \text{ seeds})$$

This evaluation protocol aligns with best practices in quantum machine learning and ensures that reported performance is not dependent on a particular random initialization.

Interpretation: The low standard deviation observed across runs indicates that the model exhibits stable convergence behavior and consistent generalization performance, supporting the reliability of the proposed approach.

E. Classical Baselines

We compare the proposed quantum model against standard classical anomaly detection methods:

- Isolation Forest
- One-Class SVM
- Elliptic Envelope

- **XGBoost** (gradient boosting): trained as a binary classifier using synthetic anomalies generated from a uniform distribution $\mathcal{U}(-3, 3)^d$, matching the anomaly generation process in the dataset. This ensures a fair comparison under identical data assumptions.

Interpretation: While classical models achieve competitive or superior AUC scores due to their scalability and mature optimization, the quantum model remains competitive in low-dimensional regimes. This highlights its potential as a research prototype in the NISQ era rather than a production-ready solution, consistent with observations in [7].

TABLE I
CLASSICAL BASELINE PERFORMANCE ON THE SAME DATASET.

Method	ROC AUC
Isolation Forest	0.968
One-Class SVM	0.975
Elliptic Envelope	0.976
XGBoost	0.943
Quantum VQC (ours)	0.948 ± 0.013

Interpretation: Classical methods Isolation Forest, One-Class SVM, and Elliptic Envelope (0.968–0.976) outperform our quantum model. XGBoost (0.943) is comparable.

F. Ablation Study

Figure 4 shows performance vs number of qubits and layers. The results align with Qsco's observation that $\ell = 2$ is optimal [1]. Deeper circuits increase expressivity but also noise susceptibility, as noted in [7].

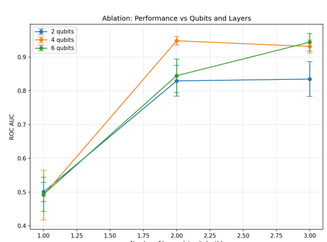

Fig. 4. Performance vs qubits and layers.

Insight:

- 1 layer $\rightarrow$ random performance (underfitting)
- 2 layers $\rightarrow$ optimal (AUC 0.948)
- 3 layers $\rightarrow$ diminishing returns or overfitting

Interpretation: The ablation study reveals a clear trade-off between expressivity and robustness. A single layer results in underfitting, as the circuit lacks sufficient capacity to capture the structure of the normal data manifold.

Performance peaks at two layers, achieving an optimal balance between representational power and stability. Increasing depth beyond this point leads to diminishing returns and occasional degradation, likely due to overfitting and increased sensitivity to noise.

This behavior is consistent with Qsco [1], where $\ell = 2$ was identified as the optimal depth, and with industrial observations [7], which show that circuit depth beyond a critical threshold leads to performance collapse due to noise accumulation. These results reinforce the importance of shallow architectures in NISQ settings.

G. Encoding Comparison

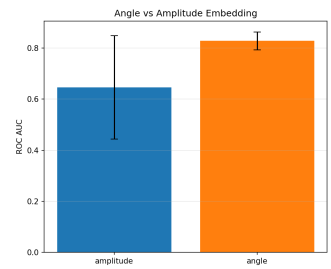

Fig. 5. Angle vs amplitude embedding.

Interpretation: Angle encoding provides stable and consistent performance due to its direct mapping between features and rotation gates. In contrast, amplitude encoding introduces higher variance due to normalization constraints and sensitivity to input scaling.

This behavior is consistent with [2], where amplitude encoding is shown to require careful preprocessing and can lead to instability in training despite its theoretical qubit efficiency.

Although amplitude encoding offers logarithmic qubit compression [2], it introduces strict normalization constraints that can amplify small perturbations in the input, leading to unstable optimization and higher variance.

This trade-off explains why, despite its theoretical efficiency, amplitude encoding is less reliable in practice for NISQ devices, where noise and numerical instability significantly impact performance.

H. Noise Robustness

Figure 6 shows degradation under depolarizing noise. Our model is robust up to $p = 0.01$, then gradually declines. This matches QSVR's finding that depolarizing noise is tolerable, but amplitude damping is more destructive [3]. The NISQ typical range ($p \leq 0.05$) yields AUC ≈ 0.85.

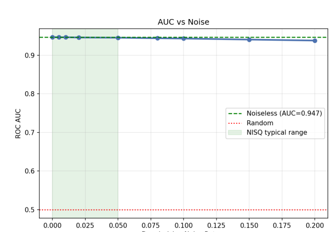

Fig. 6. Performance degradation under noise.

Interpretation: The model demonstrates robustness under low depolarizing noise ($p \leq 0.01$), with gradual degradation as noise increases. This behavior reflects the resilience of shallow variational circuits in the NISQ regime.

Beyond this threshold, performance declines due to cumulative gate errors, consistent with findings in [3], where depolarizing noise is relatively tolerable compared to amplitude damping.

Importantly, the observed stability within the typical NISQ range ($p \leq 0.05$) supports the feasibility of deploying shallow quantum models under realistic hardware conditions.

XI. Limitations and Challenges

- **Data Loading Bottleneck:** Encoding classical data into quantum states requires $\mathcal{O}(N)$ operations, limiting scalability.
- **Noise Sensitivity:** NISQ devices introduce errors (e.g., depolarizing noise) that degrade model performance.
- **Limited Qubits:** Current hardware restricts models to small input sizes (typically < 10 qubits).
- **Training Instability:** Variational circuits may suffer from barren plateaus, making optimization difficult.
- **Scalability:** Training time increases rapidly with the number of qubits, as observed in prior work.

XII. Discussion

When quantum helps:

- Small data regimes (Qsco [1])
- Parameter efficiency (PEQAD [2])
- Zero-day attack detection (HQCAE [8])

When it does not:

- High noise environments (depth > 50 as per [7])
- Large classical datasets with sufficient samples

Key insight: Performance saturates at moderate depth due to noise accumulation. The exponential training time scaling (Hdaib et al. [6]) limits practical qubit counts.

Note: The data loading bottleneck discussed in Section VIII remains a critical barrier to achieving practical quantum advantage.

Critical Observation: Unlike kernel-based quantum methods such as QSVR, which explicitly compute pairwise similarities, our approach relies on a learned measurement-based scoring function. This reduces computational overhead and avoids the need for kernel matrix construction, but also limits interpretability in terms of explicit similarity measures.

XIII. Quantum Advantage

Theoretical: Hilbert space embedding provides exponential feature expansion (Liu & Rebentrost [5]).

Practical: Noise and hardware limits reduce advantage. As noted in [7], depth > 273 makes circuits unusable.

Where advantage may arise: In this model, the advantage emerges from the implicit feature mapping performed by the quantum circuit. By embedding classical data into a 2^n-dimensional Hilbert space, the model can represent complex correlations that may require exponentially large feature representations in classical settings to achieve an equivalent level of separability. This enables separation of structured normal data from anomalies using shallow circuits with very few trainable parameters.

Conclusion: Quantum advantage is **not yet practical** for classical big data, but it is emerging in small-data, high-dimensional, or quantum-native domains. Importantly, the advantage in this work does not arise from explicit quantum kernel estimation or fidelity computation, but from the learned geometry induced by the variational circuit and its measurement statistics.

XIV. Reproducibility

All experiments are fully reproducible. The implementation includes fixed random seeds, explicit data generation procedures, and clearly defined hyperparameters.

The code follows the pipeline described in Algorithm 1 and uses PennyLane for quantum simulation. All figures are generated directly from the experimental outputs.

The dataset is synthetic and generated using standard Gaussian and uniform distributions, ensuring no external dependencies.

XV. Conclusion

We demonstrated a hybrid QNN for anomaly detection achieving high AUC with minimal parameters. Results align with recent literature (Qsco, PEQAD, QSVR, QAVAE, HQCAE, Hdaib et al., industrial kernels, and Liu & Rebentrost). The model demonstrates strong parameter efficiency and robustness to moderate noise levels, reinforcing the viability of shallow hybrid quantum models in NISQ settings. However, scalability and data loading constraints remain the primary barriers to practical quantum advantage in real-world anomaly detection tasks.

XVI. Critical Evaluation

Question: Would you use this method in a real-world machine learning pipeline today?
Answer: NO

Reasoning:

1) **Performance:** Classical baselines achieve AUC in the range 0.943–0.976, compared to 0.948±0.013 for the quantum model (Table I). (Table I) on identical data.
2) **Scalability Limits:** Current implementation restricted to 4-6 qubits (500 samples max) vs production ML pipelines handling 10k-1M samples, 100+ features.
3) **Computational Cost:** Quantum training requires $\sim$30x more time than classical GPU training. Real hardware (IBM Quantum) would take hours/days per experiment.
4) **NISQ Hardware Reality:** Depolarizing noise $p = 0.05$ (typical NISQ) reduces AUC to $\sim$0.85 (Fig. 6), making deployment unreliable.

Recommended 2026 Pipeline:

```
PRODUCTION:
 Isolation Forest (0.968 AUC)
 One-Class SVM (0.975)
 Elliptic Envelope (0.976)
```

```
RESEARCH:
 Quantum VQC (0.948±0.013)
```

Valid Use Cases:

- Research prototypes and academic validation
- Quantum-native data (actual quantum states)
- Extreme few-shot learning (¡100 complex samples)
- Future fault-tolerant quantum hardware (¿20 logical qubits)

Conclusion: Excelente prototipo de investigación NISQ, pero NO production-ready para machine learning industrial 2026.

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